

Question ID: 89541f9b

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear inequalities in one or two variables	Easy

Question

Which of the following ordered pairs (x, y) satisfies the inequality $5x - 3y < 4$?

- 1. $(1, 1)$
- 2. $(2, 5)$
- 3. $(3, 2)$

Answer

- A. I only
- B. II only
- C. I and II only
- D. I and III only

Correct Answer: C

Rationale

Choice C is correct. Substituting $(1, 1)$ into the inequality gives $5(1) - 3(1) < 4$, or $2 < 4$, which is a true statement. Substituting $(2, 5)$ into the inequality gives $5(2) - 3(5) < 4$, or $-5 < 4$, which is a true statement. Substituting $(3, 2)$ into the inequality gives $5(3) - 3(2) < 4$, or $9 < 4$, which is not a true statement. Therefore, $(1, 1)$ and $(2, 5)$ are the only ordered pairs shown that satisfy the given inequality.

Choice A is incorrect because the ordered pair $(2, 5)$ also satisfies the inequality. Choice B is incorrect because the ordered pair $(1, 1)$ also satisfies the inequality. Choice D is incorrect because the ordered pair $(3, 2)$ does not satisfy the inequality.